DATA SCIENCE V1.0

Lesson 2

Documentation: Data Science V1.0

Data Processing & Exploratory Data Analysis

Objectives

- Perform data cleaning
- Conduct EDA professionally
- Detect data leakage
- Translate analytics into business insights

Topics

- Pandas DataFrame operations
- Missing values strategies
- Outlier detection
- Feature engineering
- Correlation analysis
- Data visualization

- Business storytelling with data
- Data Preprocessing

Resources

- **Pandas**
  - https://pandas.pydata.org/docs/
- **NumPy**
  - https://numpy.org/doc/stable/
- **Matplotlib**
  - https://matplotlib.org/stable/tutorials/index.html
- **Kaggle Datasets**
  - https://www.kaggle.com/datasets
- **Book**
  - Hands-On Machine Learning – Aurélien Géron

Practice Project – Advanced EDA Report
Dataset Options

- Customer churn dataset
- Housing prices
- Credit risk dataset

Required Analysis

1. Data cleaning
   - Handle missing values (3 methods)
   - Detect duplicates
   - Type corrections
2. Exploratory analysis
   - Distribution plots
   - Correlation heatmap
   - Boxplots for outliers
   - Feature relationships
3. Feature engineering
   - Create at least 3 new features
   - Encode categorical variables
   - Normalize numeric features
4. Business Insights
   - Identify 5 actionable findings
   - Detect potential data leakage

Deliverables

- Jupyter Notebook
- Cleaned dataset
- Executive summary (1-page PDF)
- 10-minute presentation